

Stock Split Event Strategy

Technical documentation and backtest analysis

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1. Summary

The strategy trades the short term price reaction to stock split announcements in three sectors: Technology, Healthcare and Real Estate. When a split warning appears, a linear regression trained on past splits in that sector predicts the return over the next three days. The strategy goes long if the prediction is positive and short if it is negative, and closes the position after three calendar days. Between trades, the capital sits in an ETF.

Two versions were run. They differ in one line of code, the ticker of the ETF that holds idle capital: **BIL** (1 to 3 month Treasury bills) or **SPY** (S&P 500). Both runs place exactly the same 250 orders on split stocks, on 114 different companies, so the difference between them comes from the idle capital alone.

- **BIL version:** 26.5% a year, maximum drawdown 13.5%, Sharpe 1.09. Beta to the S&P 500 is 0.28 on monthly returns. This version is the cleaner measurement of the split signal.
- **SPY version:** 43.9% a year, maximum drawdown 33.4%, Sharpe 1.315. Beta is 1.25 and the drawdown is close to the S&P 500's 33.7%. About 39% of its closed trade profit comes from holding SPY, not from split trades.
- **Concentration:** in dollar terms, ten trades produce 80% (BIL) and 88% (SPY) of the profit from split trades, and about 69% of that profit comes from short positions, whose borrowing costs may not be fully reflected in the backtest.

2. The strategy

2.1 Idea

A stock split changes the number of shares outstanding without changing the value of the company. In a forward split (for example 10 for 1) each share becomes ten and the price falls proportionally. In a reverse split (for example 1 for 10) ten shares become one and the price rises. The literature reports abnormal returns around split announcements and attributes them to signalling, liquidity and retail attention. The strategy does not assume that effect exists. It tests whether a simple model of the split and of sector momentum can capture some of it after the announcement.

2.2 Signal

Each sector has its own ordinary least squares regression. The target is the return from the first open after the warning to the open three days later. There are two inputs:

- **Normalised split factor**, equal to 1 divided by the split factor. QuantConnect stores a 10 for 1 forward split as 0.1 and a 1 for 10 reverse split as 10, so the inversion gives 10 and 0.1 respectively and makes the input increase with the size of a forward split. An earlier version fed the raw factor, which reversed the meaning of the input and produced short positions after forward splits. This was fixed by the inversion.
- **Sector momentum**, the 22 day rate of change of the sector ETF (XLK, XLV or XLRE) at the time of the warning.

The models are retrained on the first day of every month using the previous four years of splits. At the end of 2025 the training sets held 177 samples for Technology, 412 for Healthcare and 48 for Real Estate. The Real Estate model rests on a small sample. A prediction of exactly zero is skipped.

2.3 Universe and sector choice

The universe contains the stocks classified by Morningstar in the three sectors. All eleven Morningstar sectors were tested on the full period, and only these three gave consistent results. The other eight had too few splits, weak or negative returns, or unstable behaviour. This screening used the same period as the reported results, so it adds selection bias, which section 5 discusses.

2.4 Execution rules

Rule	Value
Entry	Market on open order after a split WARNING event
Holding period	3 calendar days, closed with a market on open order
Position size	25% of portfolio value per trade, maximum 4 open trades
Price filter	Minimum price \$1.00
Liquidity filter	Price times volume of the latest bar of at least \$500,000 (hourly universe resolution)
Shorts	Only if the security is shortable in the backtest data
Split during a trade	Position quantity is adjusted by the split factor
Idle capital	Held in BIL or SPY. Rebalanced every day at 9:35 to the free fraction of the portfolio minus a 0.5% cash buffer, only if it drifts by more than 1%. Only the parking shares needed to fund a new trade are sold.

3. Results

3.1 Headline statistics

Metric	BIL version	SPY version	S&P 500
Net profit	420.4%	1177.8%	n/a
Compounding annual return	26.5%	43.9%	17.2%
Sharpe ratio	1.09	1.315	n/a
Sortino ratio	1.241	1.638	n/a
Probabilistic Sharpe ratio	81.1%	82.6%	n/a
Annual standard deviation	14.4%	21.6%	n/a
Maximum drawdown	13.5%	33.4%	33.7%
Drawdown recovery (days)	539	603	n/a
Calmar ratio	1.97	1.31	n/a
Alpha (QuantConnect, daily)	0.156	0.187	n/a
Beta (QuantConnect, daily)	0.015	0.976	1.000
Final equity from \$10,000	\$52,037	\$127,779	n/a
Total fees	\$1,287	\$2,102	n/a
Portfolio turnover	4.68%	5.16%	n/a

Table 1. Statistics reported by QuantConnect for each run. The S&P 500 column uses the SPY benchmark series of the same runs, price only, so its return is understated by dividends. Sharpe ratios use QuantConnect's own risk free rate model.

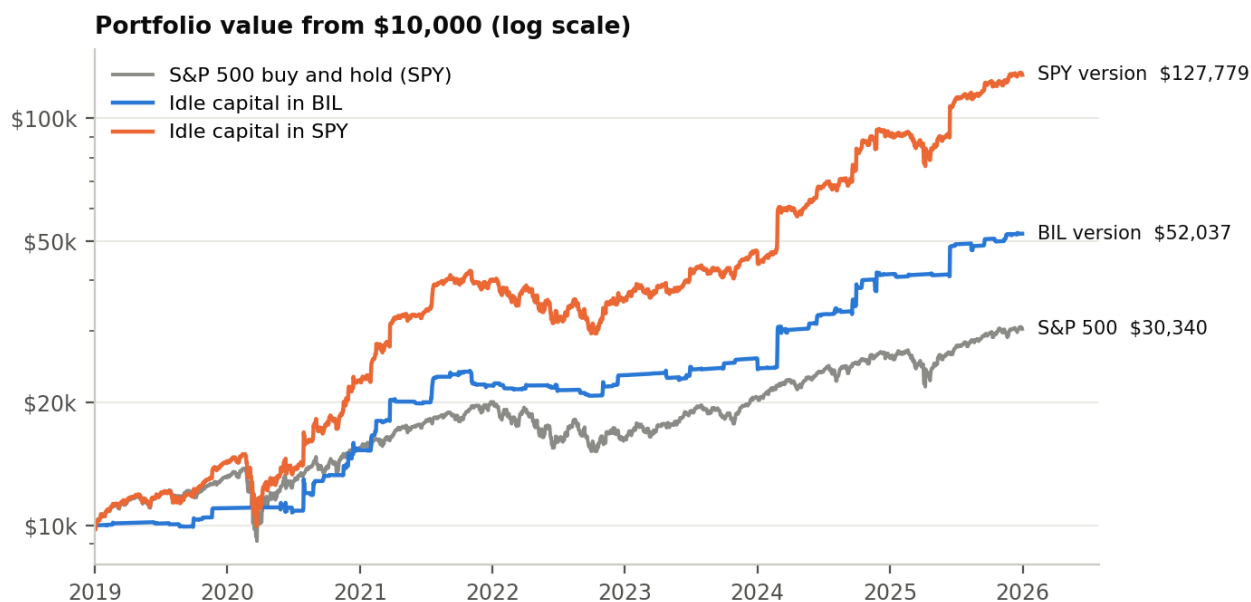


Figure 1. Portfolio value of the two versions and of the S&P 500, logarithmic scale.

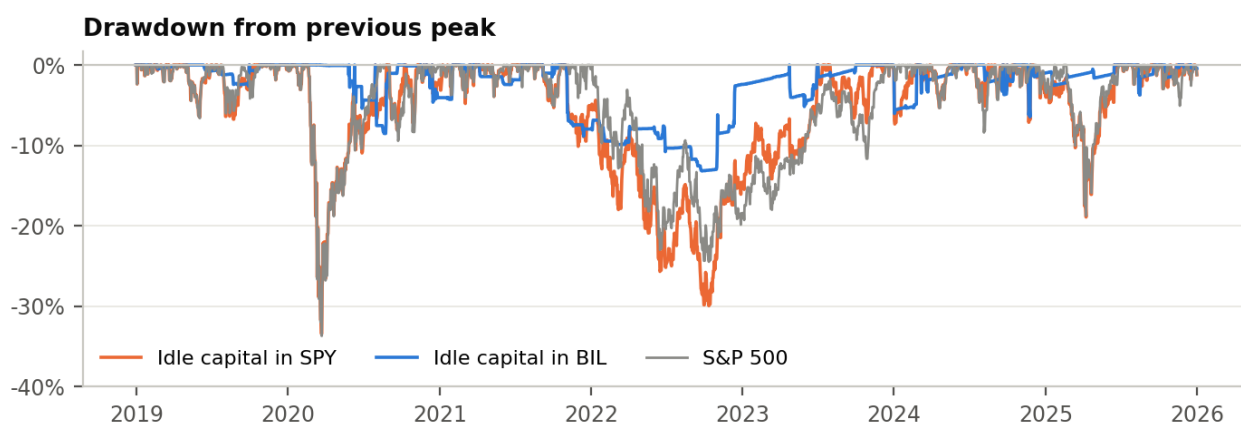


Figure 2. Drawdown from the previous peak. The BIL version shows almost no drawdown in the March 2020 crash and its worst period runs from 2021 to 2023.

3.2 Calendar years

Year	BIL version	SPY version	S&P 500
2019	10.2%	42.5%	31.2%
2020	38.7%	59.6%	18.3%
2021	44.2%	75.3%	28.7%
2022	6.0%	-9.8%	-18.2%
2023	10.0%	31.2%	26.2%
2024	60.9%	91.9%	24.9%
2025	25.7%	41.1%	17.7%

Table 2. Return by calendar year, from the daily equity curve of each run.

The BIL version finishes all seven years positive and the SPY version finishes six of seven. In 2022 the SPY version lost 9.8% against 18.2% for the S&P 500. The year 2024 alone contributes 29% (BIL) and 26% (SPY) of the cumulative log return, so the result depends on a small number of strong periods.

The models are retrained each month on data that ends before the trade, so the coefficients used in any year come from earlier data. The choices of sectors, filters and model inputs were not made that way, as section 5 explains. These calendar year returns are therefore not an independent out of sample test.

3.3 Trades

Split trades	BIL version	SPY version
Orders on split stocks	250	250
Closed trades on split stocks (QuantConnect)	134	133
Win rate	59%	59%
Mean return per trade	4.5%	4.6%
Median return per trade	1.2%	1.1%
Average winning trade	11.9%	12.1%
Average losing trade	-6.1%	-6.1%
Long trades: number, win rate	61, 51%	62, 52%
Long trades: mean, median return	2.4%, 0.2%	2.5%, 0.4%
Short trades: number, win rate	73, 66%	71, 65%
Short trades: mean, median return	6.2%, 2.0%	6.4%, 2.0%

Table 3. Trades on split stocks. Returns are profit divided by position value at entry. Returns are almost the same in the two runs because the signals are the same.

Source of closed trade profit	BIL version	SPY version
Split trades	\$36,256	\$68,720
Idle capital ETF	\$4,757	\$44,494
Total	\$41,013	\$113,214
Share from split trades	88%	61%
Share of split profit from short positions	69%	68%
Share of split profit from the ten best trades	80%	88%

Table 4. Attribution of closed trade profit. Dollar amounts are larger in the SPY version because the portfolio compounds from a higher base, so shares are the useful comparison.

The average split trade returns 4.5% but the median trade returns 1.2%, and long trades have a median of 0.2%. The profit comes from a small number of large winners, and mostly from short positions, which are closed after three days.

4. Market exposure and alpha

4.1 Single index model

Performance is measured with the single index model of Bodie, Kane and Marcus (Investments, chapter 26). Monthly returns from January 2019 to December 2025 (84 observations) are taken from the daily equity curves, converted to excess returns over a constant 4% annual risk free rate, and regressed on the S&P 500 (the SPY benchmark series of the runs).

Index model	BIL version	SPY version	S&P 500
Alpha per month	1.47%	1.65%	0.00%
Alpha annualised	19.2%	21.7%	0.0%
t statistic of alpha	2.47	2.74	n/a
Beta	0.28	1.25	1.00
t statistic of beta	2.33	10.11	n/a
R squared	0.06	0.56	1.00
Sharpe ratio (monthly)	1.14	1.32	0.81
Sortino ratio (monthly)	4.53	3.31	n/a
Serial correlation (lag 1)	-0.071	-0.024	n/a

Table 5. Single index regression on monthly excess returns. The Sharpe ratios differ from Table 1 because Table 1 uses QuantConnect's daily calculation and its own risk free rate.

- **BIL version.** Beta is 0.28 ($t = 2.33$) and the market explains only 6% of monthly variance. Annualised alpha is 19% with $t = 2.47$. Beta grows with the horizon: 0.01 on daily returns, 0.05 weekly and 0.28 monthly. The daily beta of 0.015 that QuantConnect reports therefore understates market exposure over longer horizons, and the monthly figure is the one to use.
- **SPY version.** Beta is 1.25 and the market explains 56% of monthly variance. Annualised alpha is 22% with $t = 2.74$. Beta above one comes from holding SPY with the split trades on top.
- **Serial correlation** is close to zero in both versions (-0.07 and -0.02), so there is no sign of smoothed returns, which Getmansky, Lo and Makarov (2004) associate with illiquid holdings.

4.2 Four factor style analysis

The style analysis of Bodie, Kane and Marcus (section 26.4) asks how much of the return comes from exposure to equities, interest rates, credit and the dollar. The monthly excess returns of each version are regressed on the excess returns of four ETFs: SPY for equities, TLT for long Treasuries, LQD for investment grade credit and UUP for the US dollar. The alpha is the part of the return that these four exposures do not explain.

Factor	BIL version	SPY version
Alpha per month (t statistic)	1.25% (2.09)	1.44% (2.37)
Alpha annualised	16.0%	18.7%
Equities (SPY)	0.56 (2.85)	1.50 (7.56)
Long Treasuries (TLT)	0.34 (1.36)	0.31 (1.20)
Investment grade credit (LQD)	-1.21 (-2.22)	-1.15 (-2.08)
US dollar (UUP)	-0.71 (-1.95)	-0.75 (-2.03)
R squared	0.14	0.59
Adjusted R squared	0.09	0.57

Table 6. Four factor regression on 84 monthly excess returns. Coefficients are followed by t statistics in parentheses. Excess returns use the same constant 4% annual risk free rate as Table 5.

- **Equity exposure.** The SPY coefficient is 0.56 for the BIL version and 1.50 for the SPY version. It is larger than in the single index model (0.28 and 1.25) because the credit and dollar loading on LQD, which moves with equities, offsets part of it.
- **Credit and dollar.** Both versions have a negative loading on LQD (-1.21 and -1.15, $t = -2.22$ and -2.08) and a negative loading on UUP (-0.71 and -0.75, $t = -1.95$ and -2.03), which is close to the 5% threshold. TLT is not significant in either version. The similarity between versions is expected, since the split signals are the same.
- **Overlap between factors.** LQD is correlated with TLT (0.76) and with SPY (0.67), and its variance inflation factor is 6.8. The individual loadings on LQD and TLT are therefore imprecise, and the SPY coefficient and the R squared are the more reliable results. LQD is only a proxy for the credit premium, because it mixes duration and spread, and UUP is a proxy for a trade weighted dollar index.
- **Unexplained share.** The four factors explain 14% of monthly variance in the BIL version and 59% in the SPY version. Most of the variation in the BIL version is not explained by these exposures.

After the four factors, annualised alpha is 16.0% ($t = 2.09$) for the BIL version and 18.7% ($t = 2.37$) for the SPY version. This is lower than the single index estimates, and both t statistics are still above 2, the BIL one by a small margin. They come from a strategy whose sectors, filters and inputs were chosen on the same data, so they are an upper estimate of what an unseen period would show.

5. Limitations

- **Selection on the full sample.** The three sectors, the liquidity filter and the model inputs were chosen after looking at results from the same period. Monthly retraining keeps the coefficients free of future data, but the design choices are not. A period after the last date tested is needed to remove this bias.
- **Short positions and borrowing cost.** About 69% of split profit comes from shorts. The backtest skips stocks that are not shortable, but borrowing costs may not be fully reflected, and they can be high for small Healthcare stocks. The short side may therefore be overstated by an unknown amount.

- **Concentration and sample size.** The strategy has about 134 closed trades in seven years. Ten trades account for 80% (BIL) and 88% (SPY) of split profit.
- **Execution.** Orders are market on open orders filled by the backtest, and small stocks can have wider spreads and worse fills in practice. The liquidity filter is named `avg_dollar_volume` in the code but is price times the volume of the latest bar, so it is a rough screen and not an average daily volume.
- **Capacity.** QuantConnect estimates a capacity of \$350 million for the BIL version and \$860 million for the SPY version. In each run the asset with the lowest capacity is the idle capital ETF, and the estimate scales up from a \$10,000 account. It does not measure the limit set by trading small stocks, so this document does not rely on it.
- **Drawdowns and period.** Recovery from the maximum drawdown took 539 days (BIL) and 603 days (SPY). The sample covers seven years and one rate cycle.
- **Model.** A two input linear regression will miss non linear effects. It is simple on purpose, because a model with many inputs would overfit a sample of this size.

6. Conclusion

In this backtest, a simple model of split announcements earned a positive return. The BIL version keeps almost no market exposure between trades, and it returned 26.5% a year with a 13.5% maximum drawdown and a monthly beta of 0.28. It is the better measure of the signal itself. The SPY version returned 43.9% a year, but its beta is 1.25 and its drawdown matches the S&P 500, and 39% of its closed trade profit comes from holding SPY. It is an equity portfolio with the split strategy on top. Its Sharpe ratio of 1.315 against 1.09 for BIL comes with that market exposure, in a period when the S&P 500 returned 17.2% a year.

The results do not show that the edge survives borrowing costs and realistic execution, or that it holds in a period the design choices did not see. The next test is to model borrowing costs and slippage in the backtest and to compare realised fills with simulated ones on new data.

Appendix A. Method notes

- Table 1 and Table 3 counts come from the QuantConnect result files of the two runs. The Sharpe ratio, Sortino ratio, alpha, beta and drawdown in Table 1 are QuantConnect's own calculations.
- Tables 2, 4, 5 and 6 and the figures are computed from the daily equity curves and the benchmark series in the same result files. Monthly returns use month end values, from January 2019 to December 2025.
- QuantConnect chart timestamps mark the start of a day and carry the previous close, so every series was moved back by one day to sit on its own close date before month ends were taken.
- The factor returns in Table 6 are monthly returns of dividend adjusted SPY, TLT, LQD and UUP prices from QuantConnect Research. The SPY series is identical to the benchmark series of the result files, which confirms the alignment.
- Return per trade in Table 3 is the profit of each closed trade divided by quantity times entry price, and includes fees.
- The two versions use the same code. The BIL version sets the default of the `parking_ticker` parameter to 'BIL' instead of 'SPY'.

References

Bodie, Z., Kane, A. and Marcus, A.J. (2014). Investments, 10th edition. McGraw Hill Education. Chapter 26.
 Getmansky, M., Lo, A.W. and Makarov, I. (2004). An econometric model of serial correlation and illiquidity in hedge fund returns. Journal of Financial Economics, 74, 529 to 609.
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